

IT2120 - Probability and Statistics

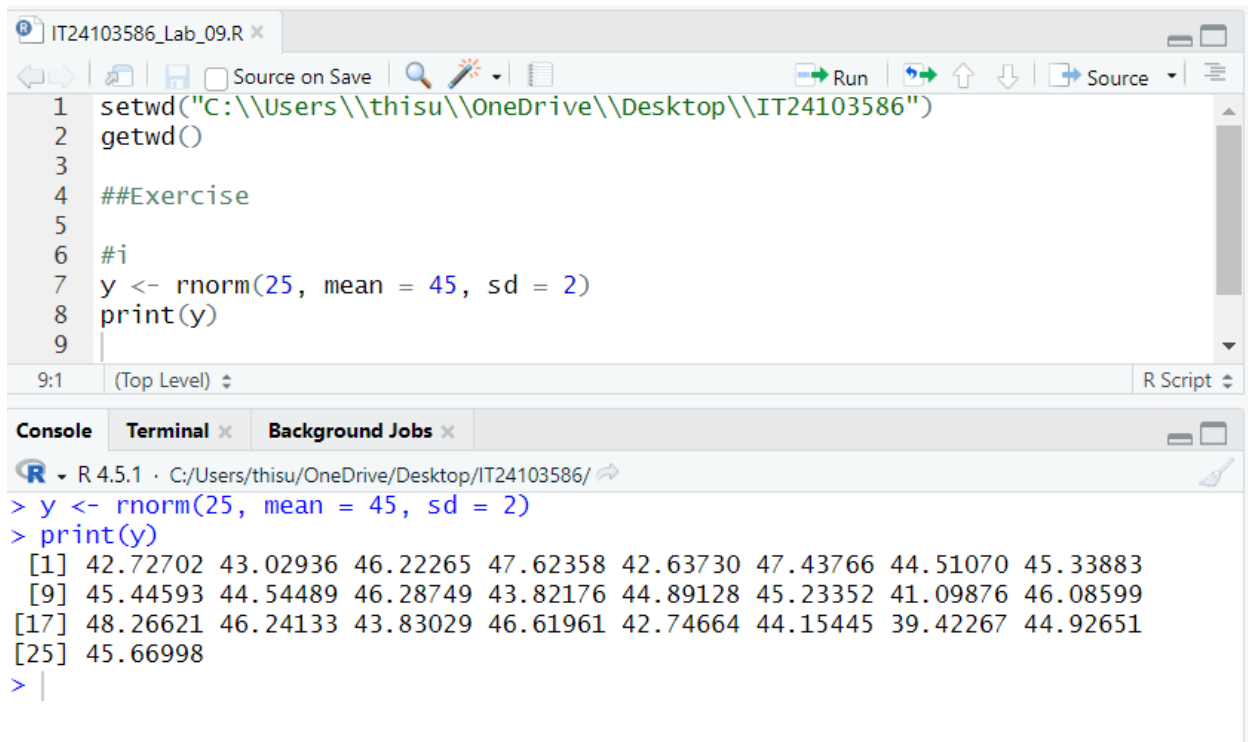
Lab Sheet 07

IT24103586 – Herath H.M.T.P

Exercise

Instructions: Create a folder in your desktop with your registration number (Eg: "IT....."). You need to save the R script file and take screenshots of the command prompt with answers and save it in a word document inside the folder. Save both R script file and word document with your registration number (Eg: "IT....."). After you finish the exercise, zip the folder and upload the zip file to the submission link.

1. Assume that the time taken to bake a batch of cookies is normally distributed with mean 45 minutes and standard deviation 2 minutes.
 - i. Generate a random sample of size 25 for the baking time.
 - ii. Test whether the average baking time is less than 46 minutes at a 5% level of significance.



The screenshot shows an R Studio window titled "IT24103586_Lab_09.R". The script editor contains the following code:

```
1 setwd("C:\\Users\\thisu\\OneDrive\\Desktop\\IT24103586")
2 getwd()
3
4 ##Exercise
5
6 #i
7 y <- rnorm(25, mean = 45, sd = 2)
8 print(y)
9
```

The console output shows the results of running the script:

```
> y <- rnorm(25, mean = 45, sd = 2)
> print(y)
[1] 42.72702 43.02936 46.22265 47.62358 42.63730 47.43766 44.51070 45.33883
[9] 45.44593 44.54489 46.28749 43.82176 44.89128 45.23352 41.09876 46.08599
[17] 48.26621 46.24133 43.83029 46.61961 42.74664 44.15445 39.42267 44.92651
[25] 45.66998
>
```

The screenshot shows an RStudio interface with a script editor and a console. The script editor contains the following code:

```
10 #ii
11 t.test(y, mu = 46, alternative = "less")
12
13
```

The console shows the output of the t-test:

```
> #ii
> t.test(y, mu = 46, alternative = "less")

One Sample t-test

data: y
t = -2.6428, df = 24, p-value = 0.007126
alternative hypothesis: true mean is less than 46
95 percent confidence interval:
 -Inf 45.61814
sample estimates:
mean of x
 44.91708
```

The RStudio window title is "IT24103586_Lab_09.R*". The console shows the R version as 4.5.1 and the working directory as C:/Users/thisu/OneDrive/Desktop/IT24103586/.